

# Types of Chemical Reactions

Answer key with worked solutions

## A. Classifying

1. Classify.

(a) **synthesis** (b) **decomposition** (c) **double displacement** (precipitation of BaSO<sub>4</sub>) (d) **single displacement** (e) **combustion**

## B. Predicting products

2. Will it react?

- (a) Mg is above Zn:  $\text{Mg} + \text{ZnCl}_2 \longrightarrow \text{MgCl}_2 + \text{Zn}$   
(b) Ag is below Cu: **no reaction**  
(c) Zn is above H:  $\text{Zn} + 2\text{HCl} \longrightarrow \text{ZnCl}_2 + \text{H}_2$   
(d) Cl is more reactive than Br:  $\text{Cl}_2 + 2\text{NaBr} \longrightarrow 2\text{NaCl} + \text{Br}_2$   
(e) Br is less reactive than Cl: **no reaction**

3. Complete and balance.

(a)  $2\text{Na} + \text{Cl}_2 \longrightarrow 2\text{NaCl}$  (b)  $\text{MgCO}_3 \longrightarrow \text{MgO} + \text{CO}_2$  (c)  
 $\text{H}_2\text{SO}_4 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$

## C. Combustion

4. Combustion.

- (a)  $\text{CH}_4 + 2\text{O}_2 \longrightarrow \text{CO}_2 + 2\text{H}_2\text{O}$   
(b)  $2\text{C}_4\text{H}_{10} + 13\text{O}_2 \longrightarrow 8\text{CO}_2 + 10\text{H}_2\text{O}$   
(c) **Carbon monoxide** (CO), a poisonous gas (or soot, C).

## D. Double displacement

5. Why one reacts.

Silver nitrate + sodium chloride gives **silver chloride, which is insoluble**, so a precipitate forms and the reaction happens:  $\text{AgNO}_3 + \text{NaCl} \longrightarrow \text{AgCl(s)} + \text{NaNO}_3$ . With potassium nitrate + sodium chloride, the possible products (KCl and  $\text{NaNO}_3$ ) are **both soluble**: no precipitate, water or gas, so no reaction.

## E. Calculations

6. Magnesium oxide.

$$\begin{aligned} & 2.43 \text{ g Mg} \times \frac{1 \text{ mol Mg}}{24.31 \text{ g Mg}} \\ & \times \frac{2 \text{ mol MgO}}{2 \text{ mol Mg}} \times \frac{40.31 \text{ g MgO}}{1 \text{ mol MgO}} \\ & = \mathbf{4.03 \text{ g MgO}} \end{aligned}$$

7. Zinc used.

$\text{Zn} + \text{CuSO}_4 \longrightarrow \text{ZnSO}_4 + \text{Cu}$ : 1 mol Zn gives 1 mol Cu.

$$\begin{aligned} & 4.86 \text{ g Cu} \times \frac{1 \text{ mol Cu}}{63.55 \text{ g Cu}} \\ & \times \frac{1 \text{ mol Zn}}{1 \text{ mol Cu}} \times \frac{65.38 \text{ g Zn}}{1 \text{ mol Zn}} \\ & = \mathbf{5.00 \text{ g Zn}} \end{aligned}$$

## F. Challenge

8. Order of reactivity.

**X > Z > Y**. A metal can only displace a less reactive metal. X displaces both others, so it is the most reactive; Z displaces Y, so Z is more reactive than Y.